

# SALMA BIBI

AI / Machine Learning Engineer | NLP • Generative AI • Agentic AI  
Multan, Pakistan | +92 301 7532504 | salmamlddev56@gmail.com | [LinkedIn](#) | [GitHub](#) | Open to Remote

## PROFESSIONAL SUMMARY

AI / Machine Learning Engineer focused on production systems — not just models. Identifies where AI actually solves the business problem, selects the right approach (ML, deep learning, or GenAI), and owns the full pipeline from data preparation to deployment. Built a full RAG pipeline, agentic AI systems, and deployed ML models via FastAPI and Docker in production.

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript, TypeScript, Dart

**AI / ML & NLP:** Scikit-learn, XGBoost, LightGBM, TensorFlow, CNN, ResNet50, Transfer Learning, Transformers, BERT, Whisper, Sentence Transformers, Prompt Engineering, RAG Pipelines

**Generative AI / LLMs:** OpenAI API, Gemini API, Groq (LLaMA), Hugging Face Inference API, LangGraph, Playwright Agent, FAISS Vector Search, Agentic Loops

**Backend / MLOps:** FastAPI, Node.js, Next.js 14, Docker, Hugging Face Spaces, Streamlit Cloud, MongoDB (ACID), MySQL, PostgreSQL

**Mobile / Edge:** Flutter, Firebase (Realtime DB, FCM), WebRTC, Android Camera2, ML Kit

## PROFESSIONAL EXPERIENCE

### AI Engineer | ClickTake Technologies, Multan, Pakistan

Apr 2026 – Present

- Built and shipped a GDPR-compliant multi-tenant SaaS repair platform (Next.js 14 + MongoDB) with JWT/RBAC across 4 permission tiers — zero cross-tenant data leakage and zero unauthorized access incidents across all production deployments.
- Improved dashboard query performance by 40% via compound MongoDB indexing, reducing average page load from ~2s to under 0.8s for concurrent enterprise tenants.
- Engineered atomic loyalty-points engine handling 500+ concurrent writes with 100% balance consistency — eliminating ~8 hrs/week of manual PII compliance work via GDPR pipeline.

### AI Automation Engineer | TheEduAssist, Pakistan

Oct 2025 – Jan 2026

- Designed and deployed NLP-driven mental health kiosk for 200+ users using Transformer-based models — achieved 87% session completion rate, the highest engagement metric at the organization.
- Built custom AI safety layer on Hugging Face Inference API preventing harmful outputs across all user interactions — maintained 100% responsible AI compliance in a sensitive healthcare context.

### ML Engineer Intern | Technocolabs Softwares (Remote)

Apr 2025 – Jun 2025

- Built ETA prediction system on 50K+ delivery records using XGBoost — reduced prediction error by 23% and improved F1-score from 0.71 to 0.89 through feature engineering on 15+ engineered features.
- Deployed real-time FastAPI inference service achieving sub-200ms latency for 1,000+ daily API calls on Hugging Face Spaces — containerized with Docker for consistent production deployment.

## KEY PROJECTS

### AI Voice Assistant for Document Q&A | End-to-End Voice RAG Pipeline

- Problem:** Users needed to ask questions about PDFs using voice — no existing tool combined STT + RAG + TTS in one free pipeline.
- Solution:** Built end-to-end voice pipeline — Whisper transcribes voice → FAISS retrieves relevant PDF chunks → Groq/LLaMA 3.3 70B generates answer → gTTS speaks it back. LangGraph orchestrates all 6 nodes. Reduced manual document querying from ~5 min to under 10 seconds per question.
- Stack:** Python, LangGraph, Whisper, FAISS, Groq/LLaMA, sentence-transformers, gTTS, Streamlit

### PDF Chat Assistant — RAG Pipeline | [Live Demo](#)

- Problem:** Users needed accurate answers from uploaded PDFs without AI hallucinating from outside knowledge — existing tools mixed document content with general AI knowledge.
- Solution:** Built complete RAG from scratch without frameworks — pdfplumber extraction, 400-word chunking with 50-word overlap, Gemini embeddings, cosine similarity search, Gemini 2.5 Flash for context-only answers. Supports 2 PDFs simultaneously. Reduced answer error rate to 0% from outside hallucination. Deployed live on Streamlit Cloud.
- Stack:** Python, pdfplumber, Gemini Embedding API, Gemini 2.5 Flash, Streamlit | [GitHub](#)

### Data Web Scraper Agent — Agentic AI | Autonomous Browser Automation

- Problem:** Manual web scraping requires human decisions at every step and breaks when page structure changes — creating bottlenecks in data collection workflows.
- Solution:** Built agentic AI system — Gemini reads page HTML, decides next action (click/type/scrape/finish), Playwright executes. Loop continues autonomously until goal complete. Eliminated manual scraping effort entirely for supported sites. Data saved as structured JSON.
- Stack:** Python, Playwright, Gemini 2.5 Flash, Streamlit | [GitHub](#)

## EDUCATION

**B.S. in Software Engineering** | Virtual University of Pakistan | CGPA: 3.35 / 4.00 | Graduated Mar 2025

Final Project (CS619): Tourism & Hotel Booking Management System